

Framework for integrating generative AI into educational tasks (MIAE), v2.1

The Framework for Integrating Generative AI into Educational Tasks (MIAE), v2.1, classifies how work is shared between the person and AI. It comprises six levels, from 0 to 5, according to the degree of autonomy and contribution of AI.

This work has been inspired by the article The Artificial Intelligence Assessment Scale (AIAS): A Framework for Ethical Integration of Generative AI in Educational Assessment. After various adaptations of the scale of the previous work, the number of modifications has become so high that I have finally decided to write this article with a new scale that **maintains little relationship** with the previously mentioned work.

Origin of the framework

Authors such as those of the aforementioned work and others such as Matt Miller focus on **plagiarism and academic deception** as the driving force behind the categorizations they propose in their works. Although the ethical aspect is very important, **we have not wanted to focus** solely on it, since it can produce a classification that is biased, unnatural and of limited application to academic integrity and ethics. Furthermore, an attempt has been made to eliminate the confusion that the previous scales have, trying to make them logically coherent in their progression.

This framework has been generalized for the integration of generative AI in educational tasks (MIAE), especially to **clarify the use made of AI in teaching work**. From this point of view it is applicable to both students and teachers. This vision has undeniable advantages

- It allows students and teachers to be placed in a **comprehensible framework**, clarifying the relationship they have with AI. This facilitates greater transparency and understanding about the use of technology in the educational environment.
- It establishes clear **boundaries** for the use of AI in the classroom, allowing educators to define the degree of AI intervention they want to allow based on educational objectives. These boundaries include considerations around plagiarism and cheating, ensuring that academic integrity is maintained.
- It offers guidance for integrating AI into the curriculum, providing a **structured approach** that helps maximize the educational benefits of AI while minimizing the risks associated with its misuse.

The framework is based on the degree of autonomy and contribution of AI in the educational process, progressing from the complete absence of AI to the autonomous generation of content by AI for educational use, supervised by humans. This approach not only addresses ethical concerns, but also offers an approach to understanding and using AI in various educational contexts, from written

assignments to projects, presentations, and teaching materials development. This integration allows teachers and students to take full advantage of AI capabilities, promoting more effective learning and teaching.

Generative AI Integration Scale

The scale consists of 6 levels. The first level, which is added for consistency, is the absence of AI, which is why it has been numbered 0.

As a conceptual map, the progression of the roles on the scale can be summarized as follows:

- **Level 0:** The person creates.
- **Level 1:** The person creates, AI reformulates.
- **Level 2:** The person creates, AI plans.
- **Level 3:** AI creates a draft, the person builds.
- **Level 4:** The person and AI co-create.
- **Level 5:** AI creates, the person supervises.

The progression describes **how the work is distributed**, not the number of words generated or the quality of the result or learning. The levels are not cumulative: a level 4 activity does not have to go through all the previous levels first. The appropriateness of each use depends on the objectives of the task.

How to classify a task and its changes

First, **what task or component is classified and who performs it** is delimited. Generating a text to analyze it and preparing its analysis are different processes. Text, images or code from the same project can also be distinguished when their development follows different processes. Units with a recognizable function in the work are distinguished, and their relationship with the whole is made explicit.

The level is determined by the contribution of AI and the role of the person in developing the result, considering the entire process. It is not fixed by the first request or by the origin of the idea. An activity may begin as a level 1 reformulation and become a level 4 co-creation. Similarly, an autonomously generated text may end up being a draft that the person significantly reconstructs, giving rise to a level 3 process. The provisional classification may change when learning about subsequent work.

In mixed processes it is advisable to indicate the levels **by phases or components**. For example, "level 2 in planning and level 1 in the reformulation of a human development", or "text without AI and illustration of level 5". If a single figure is needed, the level characterizing the development of the main product is indicated, accompanied by relevant differences. When no level sufficiently describes it, the mixed process is declared and the levels involved are specified.

The highest level of any interaction is not automatically assigned. To value a contribution, its function, its scope and how it is incorporated or transformed are observed. A **substantive** contribution modifies the content, reasoning, conceptual organization or operation of the product. It can be brief and decisive, like a

distinction that changes the conclusion. Its importance is not enough to determine the level: it also counts if the person develops it, reconstructs it, co-creates it with AI or accepts it after supervising it.

A discarded answer without influence on the work does not require raising the level of the product. It is advisable to register the query; It would not be correct to state that AI was not used at any stage. It is also not enough to check whether an answer was copied: a suggestion can influence an argument even if it is later expressed in other words. Styling corrections are still level 1, even though they greatly improve clarity.

Summary of the levels

Below we present a brief description of each level that helps to easily locate the one in which we are or are interested. Below there is a more exhaustive description with numerous **examples**, both for students and teachers.

Level 0 – Entirely human work: No AI is used at any stage. All content, ideas and structure are generated exclusively by the person using traditional resources.

Level 1 – Technical assistance from AI: AI is used for mechanical tasks (spelling correction or formatting) or to process and reformulate existing information (such as summarising, translating or drafting a text from ideas and notes provided by the person). AI contributes no new ideas, analysis or concepts to the original content.

Level 2 – AI-assisted planning and structuring: AI helps to generate initial ideas and structure the work, but the person develops all the final content. AI proposals are used for planning; the person undertakes the development and final writing.

Level 3 – Partial AI assistance (use of drafts or a “skeleton”): AI generates initial drafts or “skeletons” that already develop content, beyond reformulating material provided by the person. **The person uses this material as a starting point**, but assumes primary authorship by significantly rewriting, adapting and building on the AI-generated foundation. The interaction is mainly one-way: AI produces and the person reviews and modifies.

Level 4 – Advanced human-AI collaboration (co-creation): There is a **continuous, two-way dialogue** in which the person and AI contribute to developing the content. The person guides, challenges and refines contributions throughout the work. The result is a **closely integrated, co-created** product. The initial idea may be entirely human; the number of exchanges alone does not determine co-creation.

Level 5. Human supervision of autonomous AI: AI generates the final product autonomously based on human parameters. The human acts only as a final supervisor, checking quality and validating the product before use.

Description of each level

Each of the levels is detailed below and some of the key aspects that define them are given. We give examples for students and teachers, but in many cases they are interchangeable.

Level 0. Completely human work

There is no use of AI at any stage. All content, ideas and structure are generated exclusively by humans using traditional resources.

In this framework, no AI refers to the generative AI used to perform the task. Using the Internet or a digital tool does not in itself imply resorting to it. The following examples assume that the person performs the activity without generative assistance.

KEY FEATURES

- No generative AI is used at any time during the task.
- It depends exclusively on the knowledge, understanding and skills of the individual.
- All ideas, analysis and content are generated by the person.
- Only traditional resources are used (Internet, books, articles, notes, etc.).

EXAMPLES FOR STUDENTS

- Written exams in class.
- Supervised discussions.
- Jobs without access to AI.
- Generation of ideas without technology.
- Oral exams.
- Question and answer sessions or formative discussions between students and educators.

EXAMPLES FOR TEACHERS

- Design exam questions based on course content without using AI tools.
- Write rubrics and evaluation criteria for exams.
- Develop lesson plans using only traditional resources (Internet, textbooks, academic articles, class notes).
- Design activities and exercises that encourage the active participation of students.

Level 1. AI support

AI is used for mechanical tasks (such as spell checking or formatting) or to process and rephrase existing information (such as summarizing, translating, or organizing into tables). **AI does not provide new ideas, analysis or concepts** that were not present in the original content.

The person can provide their **ideas, arguments, notes or content** and ask AI to transform them into **an elaborate text**. This use as an advanced editor is still level 1 if AI is limited to faithfully expressing,

rearranging or reformulating them. It is not necessary that the initial material be an already written text. Providing just a topic or general idea is not equivalent to providing the content that AI will develop. Novelty is valued with respect to the material and development provided, not with respect to what the person might have in mind without having made it explicit.

The reorganization of this level is expressive: it improves the order or presentation while preserving the relationships provided. If AI proposes a plot hierarchy, new conceptual relationships or what aspects to develop, it intervenes in the planning or development of the content and the corresponding level must be examined. Changing a word to a faithful synonym is not equivalent to introducing a new conceptual distinction.

KEY FEATURES

- AI acts as a support tool that **does not generate new ideas or knowledge**, but can reorganize and reformulate existing information.
- It is limited to mechanical and reformulation tasks (summaries, creation of tables, corrections, translations, etc.).
- The resulting content **is strictly based on the ideas of the original material**, with no new analysis or concepts added by the AI.
- The person maintains full control over the content and decisions.

EXAMPLES FOR STUDENTS

- Use AI to summarize a text, where the result **is limited to condensing the ideas already present in the original**.
- AI organizes data into tables, making it easy to present without altering the original information.
- Use AI to translate a text into another language, verifying the accuracy of the translation **and that it does not add or alter the ideas of the original**.
- Grammar, punctuation, and spelling correction: The person can use AI to identify and correct grammatical, punctuation, spelling, and syntax errors in their work.
- Word Choice: AI can suggest suitable terms or synonyms to replace simpler words and phrases, helping to clarify writing.
- Provide the complete notes and arguments of a work and ask AI to write them clearly, checking that it preserves its meaning and does not add new explanations or conclusions.

EXAMPLES FOR TEACHERS

- Organize data in tables or graphs for presentations, making it easy to view information without altering the original content.
- Grammatical, punctuation and spelling correction in documents and educational materials, improving the quality of the text.
- Suggestion of appropriate terms or synonyms to improve the clarity of study guides and written materials.
- Assistance in preparing lesson plans, summarizing and organizing information provided according to the objectives and criteria already established by the teacher.

- Extract lists of key terms and concepts present in the material provided for exams and study guides, following the teacher's selection criteria.
- Support in the revision of students' work and assignments, using AI to identify errors of expression and provide linguistic corrections, without evaluating or developing the disciplinary content.

Level 2. AI-assisted planning and structuring

AI helps in the initial generation of ideas and structuring the work, but all the final content is developed by the human. Their proposals are used to plan; the person carries out the development and final writing.

This level includes proposing a structure, a plot sequence or a development plan. The suggested ideas and organization can guide the outcome; AI does not elaborate the arguments, analyzes or solutions that constitute it. If it is re-planned during work, that help remains level 2 as long as it retains this preparatory function.

If AI helps to plan, the person then develops all the content and finally AI reformulates it without adding ideas, **level 2 in planning and level 1 in reformulation** are combined. If, in addition to planning, AI develops the content, levels 3, 4 or 5 must be examined depending on the person's subsequent role.

KEY FEATURES

- AI is mainly used for initial idea generation and content structuring.
- It helps in the design and planning of the work, but not in the creation of the final content.
- The human uses the AI's suggestions as a starting point, but develops the work himself, applying his own judgment and knowledge.
- The final product does not incorporate AI development or writing, although suggested ideas and structures can guide the work.
- The person is responsible for critically evaluating, selecting and developing the ideas generated by the AI, ensuring that the final product is the result of their own analysis and understanding.
- The contribution of AI is limited to the preparation of the work or its parts, while the execution and detailed development are carried out entirely by the human.

EXAMPLES FOR STUDENTS

- **Initial phases of a paper.** Use of AI to generate ideas and create an outline or index for an essay, but the final writing is done without AI.
- **Brainstorming.** Students can use AI to generate ideas or possible approaches to solving problems. These ideas are discussed, filtered and refined; The students then develop the solutions.
- **Schemes.** People can use AI to create outlines of their work as a planning tool. These schemes serve as a guide for the human development of work; Your organization can be reflected in the result. If the assignment consists of delivering the scheme itself as a final product, its generation must be classified as such and not as mere planning.

- **Research assistance.** AI can suggest topics, areas of interest or useful sources for research, also through search engines.

EXAMPLES FOR TEACHERS

- **Develop a lesson plan.** Use AI to generate ideas and create an outline or table of contents for a lesson plan, but the final development of the plan is done without AI.
- **Schemes.** Create structured outlines for study units or course programs using AI, as a guide for the teacher to later develop the units or program.
- **Assistance in academic research.** AI can suggest useful topics, areas of interest, or sources for educational research, making it easier to find references.
- **Generation of ideas for class projects and activities.** AI gives ideas for activities that are then created, adapted and refined by the teacher.

Level 3. Partial AI support (use of drafts or “skeleton”)

AI generates initial drafts or “skeletons” that already develop content, beyond reformulating provided material. The human uses this material as a starting point, but assumes primary authorship, significantly rewriting, adapting and building on the foundation generated by the AI. The interaction is mainly **one-way**: AI produces and the human reviews and modifies.

A “skeleton” at this level contains an initial development that the person reconstructs; an index that only guides the work corresponds to level 2. Meaningful reconstruction affects the content, reasoning or functioning of the result. It is not enough to select an answer or make stylistic adjustments. There can be subsequent queries to AI without the process being level 4, if the person continues to do the main development themselves.

KEY FEATURES

- The interaction is mainly one-way: AI produces, the human reviews and modifies.
- The human uses AI to generate drafts, proposals or “skeletons” that serve as a starting point.
- The use of AI focuses on specific tasks where it provides efficiency, allowing the human to focus on aspects that require other skills.
- The person maintains significant control over the process and the result, assuming final authorship of the content.

EXAMPLES FOR STUDENTS

- **Writing sections of a report.** AI can write an initial draft of a section. Students review this section, making modifications, adding their own ideas, and ensuring that the final content reflects their personal understanding and analysis.

- **Data analysis.** AI carries out preliminary analysis of a data set and generates graphs or tables and interprets them. Students review and significantly rework that interpretation, incorporate their conclusions, and construct their own report on the implications of the data.
- **Writing questions for a questionnaire.** AI generates the questions of a questionnaire and the person selects and significantly reworks them, reviewing their approach, the contents they evaluate and their suitability to specific needs.
- **Presentation creation.** AI generates a draft of a presentation and the students develop and reorganize their arguments, incorporate their own explanations and adapt the materials.

EXAMPLES FOR TEACHERS

- **Generation of teaching resources.** AI creates drafts of teaching resources, such as reading guides or supplementary materials. The teacher reviews, adapts and develops these resources to ensure that they are suitable for the specific needs of their students and the curriculum.
- **Design of learning activities.** AI proposes a series of activities for a thematic unit. The teacher evaluates each activity, modifies some to make them more interactive or relevant to their specific context, and adds new activities based on their experience and knowledge of their students' learning needs.
- **Development of practical cases.** AI generates a list of possible use cases based on the key concepts of the course. The teacher selects, reviews and enriches the case, adding real context and analysis questions that encourage critical thinking and the practical application of the acquired knowledge.
- **Exam creation.** AI proposes a set of questions for an exam. The teacher selects the most appropriate ones, modifies others to adjust their difficulty or focus, and adds his or her own questions to ensure a complete and balanced assessment of students' knowledge and skills.

Level 4. Advanced human-AI collaboration (co-creation)

There is a **continuous, two-way dialogue** in which the person and AI contribute to the development of the content. The person actively guides AI during this elaboration, contrasting and refining the contributions through the conversation. The result is a **tight, co-created fusion** of their work, where AI acts as an active collaborator.

The initial idea may be completely human. What defines co-creation is that dialogue intervenes in its development through contributions from both. The person can contribute with arguments, objections, criteria and content decisions expressed in the conversation; You do not need to directly write a part of the final text.

Repetition of instructions, acceptance of proposals or requests for style changes are not enough on their own. Nor is a conversation limited to preparing ideas that the person will later develop without AI. There must be shared elaboration of the content of the task, with human direction and judgment during that development.

KEY FEATURES

- There is a continuous, two-way dialogue between the human and the AI, with iterations in which both develop and refine the content.
- The human maintains a crucial role in directing, refining and validating the work, ensuring the overall consistency and quality of the final product.
- Human creativity and critical judgment are applied to improve, personalize and give coherence to AI result.
- The person acts as coordinator and director of the project, making critical decisions about the structure, content and presentation of the work.

EXAMPLES FOR STUDENTS

- **Research project.** The student and AI collaborate in an iterative process. AI helps search for sources, propose text drafts, and create graphics. The student dialogues with AI to adjust said content, requesting corrections, extensions and changes in focus to ensure its accuracy and relevance.
- **School work.** The student provides the data necessary to do the work. AI writes a report that is examined by the first, who indicates the corrections to be made. A dialogue is established where students contrast interpretations, provide arguments and decide which explanations to revise or expand with AI.
- **Mathematical problem solving.** AI provides detailed steps and solutions for advanced problems. The student contrasts the steps, proposes reasoning and develops alternative solutions with AI, justifying the decisions that he incorporates into the final answer. Asking for explanations to understand a solution already given is not enough, in itself, to classify its elaboration as co-creation.
- **Creation of a historical podcast.** A student begins by providing AI with their general idea. AI generates an initial outline. The student reviews this outline, requests changes, and asks AI to expand on certain aspects. AI produces draft scripts that the student edits. They collaborate in the creation of notes and complementary materials. This process continues until the podcast reaches a high level of quality.
- **Mathematical model applied to climate change.** The student describes the problem to the AI. AI suggests approaches and the student asks it to develop them. Ask AI for a draft of the model. The student reviews the model, adjusts variables and incorporates additional factors with the help of AI. AI generates visualizations and the iterative process continues to refine the model.

EXAMPLES FOR TEACHERS

- **Planning of teaching units.** The teacher and AI iteratively design a complete unit. The teacher asks for objectives, AI generates them, the teacher asks to adjust them for their context, AI suggests activities and the teacher asks to modify them, in a constant dialogue.
- **Development of evaluations.** The teacher collaborates with AI to generate questions, rubrics and evaluation criteria. Through dialogue, refine the difficulty level, request question variations, and adjust rubrics until they align with course objectives.
- **Development of interactive materials.** The teacher directs AI to create drafts of simulations or games. Through iterations, the teacher reviews, requests adjustments in mechanics or content, and complements the materials, co-creating an engaging and effective resource.

- **Development of your own idea.** The teacher starts with an idea that is completely his or her own and asks for help to write it. During the dialogue, AI proposes to break down concepts, explore consequences, introduce distinctions or delve into certain aspects. The teacher contrasts these proposals, provides his reasons, accepts some, rejects others and guides their development. The contributions prepared jointly influence the final content. The process may have started at level 1 and moved to level 4; The change depends on this shared elaboration, not on the origin of the idea.

Level 5. Human supervision of autonomous AI

AI generates the content or product autonomously, with the human acting as a supervisor. The human sets the initial parameters, but AI performs the work with minimal or no intervention during the process.

The role of the human is to validate the final product for use or delivery, ensuring that it meets the initial requirements.

Autonomy refers to the development of the commissioned content or product. Providing the topic, data, or requirements does not preclude this level if AI performs that development and the person simply supervises it. Subsequent minor tweaks do not in themselves constitute level 3 reconstruction or level 4 co-creation. On the other hand, writing that only reformulates already developed human content corresponds to level 1.

KEY FEATURES

- AI generates the content independently, based on the initial parameters established by the person.
- Human intervention during the generation process is minimal or absent, limited to the initial configuration and final review.
- The content generated by AI is the final product itself, which is directly validated, approved and used by the human.
- The human acts as a director or validator who approves the result, unlike Level 4 (where they are a co-creator) or Level 3 (where they are a builder).

In the following examples, **level 5 corresponds to the generation of the resource**: the story, the report, the historical text or the dialogue. Subsequent analysis, comparison or discussion are different tasks and are classified according to how they are performed. If the teacher provides the resource and the student analyzes it without AI, the student's task can be level 0. If the student himself orders the resource and then analyzes it, it is advisable to describe both parts; the whole is not a job without AI.

EXAMPLES FOR STUDENTS

- **Literary analysis:** AI generates a story or short story on a specific literary topic or genre. The student uses this story to carry out a literary analysis, identifying themes, motifs, stylistic devices and narrative structures. The student's final work is not the modification of the story, but the critical analysis of the text generated by the AI.
- **Scientific Data Analysis:** AI produces a comprehensive report based on a set of scientific data, such as the results of an experiment or simulation. The student uses this report to perform critical

analysis, discuss methodology, interpret results, and evaluate conclusions. The report generated by AI is treated as a source document on which the student performs his analysis.

- **Historical Case Study:** AI creates a detailed description of a historical event, including narratives, key dates, and actors involved. The student uses this text to make a comparison between this event and another studied in class, developing a comparative essay that analyzes the similarities and differences in historical contexts.
- **Philosophical Discussion:** AI composes a philosophical dialogue between two characters on a contemporary ethical issue. Students use this dialogue as the basis for a class debate, where they discuss each character's positions, analyze their arguments, and develop their own philosophical reflections based on the text.

EXAMPLES FOR TEACHERS

- **Text analysis in class:** AI generates a set of expository texts about different scientific theories or mathematical concepts. The teacher uses these texts as material for students to carry out comprehension analysis, discuss in groups or answer specific questions about the concepts presented in the generated texts.
- **Study of simulated historical documents:** AI produces simulated speeches or documents of historical figures in different contexts. The teacher uses these documents in class so that the students analyze them as an exercise of interpretation, expressly identifying them as simulations and without attributing them value as an authentic historical source, developing essays or discussions based on the generated content.
- **Data Interpretation Practice:** AI creates a series of technical or scientific reports that include data analysis, graphs, and conclusions. The teacher uses these reports in class exercises where students practice interpreting data, critique the methodology used, and develop their own conclusions based on the information presented.
- **Preparation for a debate:** AI writes texts that represent different positions on a controversial topic. The teacher uses these texts to prepare students for a debate, where each group analyzes and defends the position presented by the AI, developing arguments from the base text.

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